Phase-Associative Memory: Sequence Modeling in Complex Hilbert Space

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(Dated: April 24, 2026)

We present Phase-Associative Memory (PAM), a recurrent sequence model in which all representations are complex-valued, associations accumulate in a matrix state $S_t \in \mathbb{C}^{d\times d}$ via outer products, and retrieval operates through the conjugate inner product $K_t^* \cdot Q_t/\sqrt{d}$. Across a five-scale ablation from 5M to 100M parameters on WikiText-103, PAM and a matched real-valued ablation (RPAM) both follow clean power laws in validation loss and perplexity, with PAM's slope steeper in both. The slightly steeper slope in both loss and perplexity suggests that the complex-valued architecture would outperform its real-valued ablation at modest parameter sizes (∼1–10B), at projected losses (∼2 nats) comparable to those of transformer-based language models on web-scale corpora at similar scales. We trace the experimental path from vector-state models, where holographic binding fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations, to the matrix state that resolves it. The differential scaling is consistent with the interpretation that the real-valued model is approximating, with classical machinery, a structure that the complex-valued model represents natively, and connects to recent empirical evidence that semantic interpretation in both humans and large language models exhibits non-classical contextuality. The ∼1.69 nat irreducible-loss floor characterized in scaling-law analyses of real-valued architectures need not bound a model that represents this non-classical structure natively.

INTRODUCTION

The assumption that a system's constituents can be analyzed independently of one another and of the conditions under which they are observed has been a foundational premise of empirical science since the seventeenth century [1, 2]. Scholastic metaphysics, notably Aquinas [3], treated nature as composed of distinct substances knowable in isolation [4]; early modern figures from Galileo [5] and Bacon (*Novum Organum* [6]) through Descartes [7] to Newton's *Principia* [8] sharpened a picture in which states and causes admit description without essential reference to the observer.

For more than two centuries classical physics reinforced separability and determinism as features of the world. Quantum mechanics challenged that picture: Heisenberg preserved pre-existing properties disturbed by measurement, whereas Bohr held quantities indeterminate prior to measurement. Einstein, Podolsky, and Rosen argued for definite properties of separated systems [9]; separability was central [10, 11]. Bell's theorem [12, 13] and experiments [14, 15] favored Bohr [16]: no pre-existing, context-independent values reproduce the correlations. The Kochen–Specker theorem [17] extended contextuality to single systems. Later tests closed loopholes [18–21]; quantum information [22–24] showed inseparability as a resource, with the Tsirelson bound [25] quantifying quantum-over-classical advantage.

The study of language has developed under the same separability assumptions, though the connection is rarely made explicit. The principle of compositionality, which holds that the meaning of a complex expression is determined entirely by the meanings of its parts and the rules by which they combine [26, 27], treats semantic content as a property of linguistic constituents that can be analyzed independently of the interpreter and the context of interpretation. Whether this principle is adequate as a foundation for the study of meaning has been contested on philosophical grounds for more than a century [28–30], but the computational study of language has largely proceeded as if it were settled. Zellig Harris showed that the distributional properties of words in a corpus could serve as a proxy for their semantic relationships [31], and this insight carried through the twentieth century into latent semantic analysis, word embeddings [32, 33], and ultimately large language models [34–36]. At each stage, the underlying computational assumption has been the same: words have meanings that can be represented as fixed points in a real-valued vector space, and the task of a model is to learn where those points are and how they compose. The transformer architecture [37, 38] is the most successful embodiment of this program.

Transformer-based large language models have largely succeeded in passing the well-established benchmarks of artificial intelligence, including conversational assessments that would have been considered definitive demonstrations of language understanding a decade ago [36]. However, their adoption in domains that require guaranteed reliability has been hindered by persistent difficulties, most prominently hallucination [39, 40] and susceptibility to prompt injection [41, 42], which have resisted solution despite substantial engineering effort across architectures and scales. The improvements in capability that once accompanied increases in model size and training data [43, 44] have plateaued [45, 46], and the frontier of the field has shifted toward test-time compute [47], chain-of-thought reasoning, and agentic iter-

ation as strategies for navigating the space of possible responses rather than producing the correct one directly. This shift is consistent with the observation that the informational burden of disambiguating a semantic expression grows superlinearly with its complexity [48], making the recovery of a single intended meaning from an expression of even moderate depth an intractable problem in the sense of relevance realization [49, 50]: the system cannot determine what is relevant from the input alone and must instead explore. Efforts to understand these systems through mechanistic interpretability, which attempts to decompose the internal representations of neural networks into individually meaningful components, have encountered difficulties that appear to be structural rather than merely technical. Sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features have been shown to be "neither selective nor independent" when used for steering [52], and a recent review with twenty-nine co-authors described the field's foundational concepts as "not yet established" and its status as "pre-paradigmatic" [53]. Theoretical work has demonstrated an exponential gap between the complexity of representing features in superposition and computing with them [54], and sparse autoencoders have been proven to fail to recover ground-truth features except under conditions of extreme sparsity [55]. The difficulty of producing reproducible, meaningful decompositions of neural network representations mirrors the experience of cognitive neuroscience, where decades of functional neuroimaging have shown that localized brain-behavior associations require sample sizes orders of magnitude larger than most studies have used [56], that seventy independent teams analyzing the same fMRI dataset reach substantially different conclusions about which brain regions are involved [57], and that inferring cognitive processes from regional activation is logically unreliable because brain regions participate in many functions simultaneously [58, 59]. In both cases, the assumption that the system can be understood by decomposing it into separable, localizable components has produced results that do not replicate.

In physics, the distinction between a decomposition that has not yet been found and one that cannot exist in principle, because the underlying properties are fundamentally indeterminate prior to measurement, is precisely what Bell's theorem was designed to settle, and the same framework can be applied to semantic interpretation. When the CHSH test is applied to human semantic judgments, the correlations between interpretations produced under different contextual framings violate the classical bound [60–65], and when the same tests are applied to large language models trained on text that human cognition produced, the violations persist across four orders of magnitude in parameter count, with the distributional character of the contextuality orthogonal to every standard benchmark tested [48, 66]. Sheaf-theoretic analysis of BERT's internal representations has identified over 77,000 instances of contextuality at the level of the embeddings themselves [67, 68]. The non-separability is not confined to the behavioral outputs of these systems; it is present in the geometry of their learned representations [69, 70]. Reconsidered under the premise that meaning is indeterminate prior to the act of interpretation and that natural language is semantically degenerate [48], it necessarily follows that hallucinations and jailbreaks are not anomalies to be eliminated but commonplace consequences of a system that interprets rather than retrieves [66]. If the correlational structure of language is genuinely non-classical, the natural mathematical framework for describing it is the same one that was developed for quantum mechanics: a complex Hilbert space in which states carry phase, similarities are computed through the conjugate inner product, and interference between components is an intrinsic property of the algebra rather than a behavior that must be learned. Large language models built on real-valued representations and softmax attention may functionally replicate this structure, but they do so in the way that any classical simulation of a quantum system does: by using enough parameters to project the complex-valued correlations onto a real-valued space, at a cost in capacity and efficiency that grows with the complexity of the structure being represented.

The representation of signals in complex form has a long history in engineering and physics. Gabor [71] introduced the analytic signal in his theory of communication, and Oppenheim and Lim [72] demonstrated that phase carries more structural information than magnitude in both images and audio [72]. The geometric phase discovered independently by Pancharatnam in optics [73] and Berry in quantum mechanics [74] showed that phase relationships encode topological properties of the space traversed by a system, information that is lost entirely when the representation is projected onto real-valued magnitudes. Complex-valued neural networks have been developed along these lines for decades [75–79], and the holographic reduced representations introduced by Plate [80] demonstrated that complex multiplication and conjugation provide a natural algebra for binding and retrieving associations [81, 82]. Danihelka *et al.* [83] incorporated this algebra into an LSTM with complex-valued cell states, and Ramsauer *et al.* [84] showed that the mathematical structure underlying softmax attention is a modern Hopfield network [85, 86] whose linear variant is the fast weight programmer [87, 88]. None of these efforts, however, has produced a complete language model that operates in complex space from embedding through retrieval to output at a scale where comparison with conventional architectures is meaningful.

Separately, the development of efficient alternatives to

the transformer's attention mechanism has produced a body of work that provides the architectural scaffolding for such a model. The transformer [37] computes attention as a softmax-normalized dot product between real-valued projections of the input, an operation that is powerful but quadratic in sequence length and requires a key-value cache that grows linearly during inference. Removing softmax yields a recurrence with matrix state $S_t = S_{t-1} + V_t K_t^\top$ [89], which Schlag *et al.* [88] showed is equivalent to the fast weight programmer introduced by Schmidhuber [87], an associative memory that accumulates associations via outer products and retrieves via matrix-vector product. Subsequent work has refined this structure in various ways: RetNet [90] adds exponential decay, GLA [91] introduces data-dependent gating, DeltaNet [92] replaces additive accumulation with a delta rule, and GateLoop [93] uses complex-valued gates. From the state-space model side, the Linear Recurrent Unit [94] established the importance of complex-valued diagonal recurrences for stable long-range modeling, Mamba [95] introduced input-dependent selection, Griffin [96] validated gated linear recurrence at scale, and Mamba-2 [97] proved the formal equivalence between structured SSMs and linear attention. From the LSTM lineage, mLSTM [98] independently arrives at the same matrix-state recurrence, and RWKV [99, 100] has demonstrated this family at up to 14B parameters. With few exceptions, these models operate in real-valued space. Ramsauer *et al.* [84] showed that softmax attention implements a modern Hopfield network, and the linear variant of this associative memory is precisely the fast weight programmer that the matrix-state models generalize.

The tradition of operational quantum logic, beginning with Birkhoff and von Neumann's observation that the propositions of quantum mechanics form a non-Boolean lattice [101] and developed through the work of Mackey, Piron [102], and Foulis and Randall [103, 104], spent half a century establishing that any system whose observables are contextual requires a non-Boolean algebraic structure, and that the natural home for this structure is a complex Hilbert space with the conjugate inner product. The point is not that such systems are doing quantum physics but that the algebra of complex-valued superposition and interference is the correct formalism for contextual measurement regardless of substrate. Transformers already capture the non-classical correlational structure of language, as the Bell violations demonstrate, but they do so by projecting it onto real-valued space using enough parameters to approximate the complex-valued correlations. Phase-Associative Memory (PAM) takes the matrix-state recurrence shared by the lineages described above and moves it into the space that operational quantum logic identifies as native to contextual systems. The state, keys, values, and queries are all complex-valued, and retrieval uses the conjugate inner product $K^* \cdot Q$ rather than the standard dot product, so that the selectivity of retrieval depends on the phase alignment between stored and queried representations.

The architecture emerged through a series of experiments in which each failure was informative. Early versions introduced tokens in complex phase space but destroyed phase information by passing representations through real-valued nonlinearities; correcting this with phase-preserving primitives materially improved results. A subsequent attempt to inject holographic key–value bindings into a vector-state SSM caused a regression in perplexity, because multiple bindings superposed in a single d-dimensional vector interfere destructively with the classical $O(1/\sqrt{n})$ capacity degradation [80]. PAM resolves this by upgrading the state from $\mathbb{C}^d$ to $\mathbb{C}^{d\times d}$, providing $O(d^2)$ associative capacity per head. The reported configuration interleaves channel mixing and PAM in each of 16 blocks with complex rotary position embeddings [105] on queries and keys, and admits a dual computational form that is $O(T^2)$ for parallel training and $O(1)$ per token for recurrent inference with no KV cache.

PAM trains stably on WikiText-103 across the 5M to 100M parameter range we explored, with the matrix-state recurrence and complex parameterization carrying the entire signal path without recourse to softmax attention. A preliminary comparison at ∼100M parameters places PAM within ∼10% of a matched dense transformer baseline; broader evaluation across the contemporary transformer design space is left to future work.

METHOD

The model consists of a complex-valued embedding layer, 16 identical blocks, and a tied complex output head. Each block applies channel mixing via a ComplexGatedUnit (CGU) followed by sequence mixing via a Phase-Associative Memory (PAM) layer, both with residual connections and learned scaling. All operations in the main signal path are complex-valued and phase-preserving; gates and decay parameters use real-valued projections over magnitude features, but the primary data path never converts complex representations to real-valued intermediate forms.

Complex quantities are represented as tensors with shape $[\ldots, d, 2]$, implementing $\mathbb{C}^d$ in split-real form. The complex linear map, given weight matrices $W_r, W_i \in \mathbb{R}^{m\times n}$, computes $y_r = W_r x_r - W_i x_i$ and $y_i = W_i x_r + W_r x_i$. The activation function is modReLU, $\mathrm{modReLU}(z) = \mathrm{ReLU}(|z| + b) \cdot z/|z|$ with learned bias b, which thresholds magnitude while leaving phase untouched. Normalization is RMS normalization applied to magnitudes with phase preserved: $\mathrm{ComplexNorm}(z) = s \cdot (|z|/\,\mathrm{RMS}(|z|)) \cdot z/|z|$ with learned scale s. The channel mixing layer (CGU) is a SwiGLU-style gating block

in complex space:

$$\mathrm{CGU}(z) = W_{\mathrm{down}}\big(\mathrm{gate}_{\mathrm{phase}} \odot \mathrm{modReLU}(W_{\mathrm{up}} z) \cdot \sigma(|W_g z|)\big) \tag{1}$$

where the gate magnitude $\sigma(|W_g z|)$ controls how much signal passes and the gate phase controls what rotation is applied. Each of 16 blocks applies CGU then PAM with residual connections and learned scaling:

$$\tilde{z}^{(l)} = z^{(l-1)} + \alpha^{(l)}_{\mathrm{CGU}} \cdot \mathrm{CGU}_l(\mathrm{ComplexNorm}(z^{(l-1)})), \tag{2}$$

$$z^{(l)} = \tilde{z}^{(l)} + \alpha^{(l)}_{\mathrm{PAM}} \cdot \mathrm{PAM}_l(\mathrm{ComplexNorm}(\tilde{z}^{(l)})) \tag{3}$$

where $\alpha^{(l)}_{\mathrm{CGU}}$ is initialized to 1.0 and $\alpha^{(l)}_{\mathrm{PAM}}$ to 0.1. Logits are computed via a tied complex inner product with the embedding table: $\mathrm{logits} = z_{\mathrm{out},r} \cdot E_r^\top + z_{\mathrm{out},i} \cdot E_i^\top$.

PAM replaces both the recurrent backbone and the attention mechanism with a single module whose operations correspond directly to the quantum semantic framework described in [48]. In that framework, a semantic expression S_E is represented as a state vector $|\psi_{S_E}\rangle = \sum_i c_i |e_i\rangle$ in a complex Hilbert space, where the complex coefficients c_i carry phase information with no classical analogue, and interpretation is the application of a Hermitian operator whose eigenstates represent possible meanings. PAM implements this structure computationally: tokens are embedded as complex vectors, associations between them are accumulated in a complex matrix state via outer products, and retrieval is the projection of a query onto the accumulated state through the conjugate inner product, the same operation that computes $P(m_i) = |\langle e_i|\psi_{S_E}\rangle|^2$ in the quantum semantic framework.

Each PAM head h maintains an independent complex matrix state $S_t^{(h)} \in \mathbb{C}^{d\times d}$, where d is the head dimension. Unlike attention heads, which compute independent dot-product similarities over a shared representation, PAM heads are parallel associative memory banks: each accumulates its own key-value associations and retrieves independently, so the H heads collectively maintain H separate $d \times d$ memory matrices. The total state capacity across H heads is $H \times d^2$ complex values per layer ($6 \times 64^2 = 24{,}576$ in our configuration). The input $x_t \in \mathbb{C}^D$ is projected into queries, keys, and values via a single complex linear map:

$$[Q_t; K_t; V_t] = W_{\mathrm{QKV}} x_t \quad \Rightarrow \quad Q_t, K_t, V_t \in \mathbb{C}^{H\times d}. \tag{4}$$

Complex rotary position embeddings [105] are applied to Q and K by multiplying each element by a precomputed unit-magnitude factor $e^{im\theta}$, encoding absolute position in phase while leaving magnitudes unchanged; in the conjugate product $K_i^* \cdot \tilde{Q}_t$ the dependence on position difference $(m-n)$ yields relative position structure. Retrieval uses the scaled query $\tilde{Q}_t := Q_t/\sqrt{d}$.

The decay rate γ_t controls how quickly the state forgets and is computed from the input as $\gamma_t = \exp(-\mathrm{softplus}(W_{dt} \cdot \mathrm{concat}(x_{t,r}, x_{t,i}) + b_{dt}))$, where b_{dt} is initialized to -4.0 for slow initial decay. A learned protect gate $p_t = \sigma(W_p \cdot |x_t| + b_p)$ with $b_p = -3.0$ modifies the effective decay:

$$\gamma_t = e^{-dt_t} \cdot (1 - p_t) + p_t, \qquad V'_t = V_t \cdot (1 - p_t). \tag{5}$$

When $p_t \to 1$ the state is frozen and new values are suppressed; when $p_t \to 0$ the decay proceeds normally. The state then evolves as:

$$S_t = \gamma_t \cdot S_{t-1} + V'_t \otimes K_t^* \tag{6}$$

where $\otimes$ denotes complex outer product and K_t^* is the complex conjugate of the key. Retrieval computes $Y_t = S_t \tilde{Q}_t$, which expands to:

$$Y_t = \sum_{i\le t} \left(\prod_{j=i+1}^{t} \gamma_j \right) \left(K_i^* \cdot \tilde{Q}_t\right) V'_i. \tag{7}$$

The conjugate inner product $K_i^* \cdot \tilde{Q}_t$ determines retrieval strength through phase alignment: associations whose keys are phase-coherent with the query are retrieved strongly while phase-incoherent associations are suppressed, without softmax normalization. When the keys are mutually orthogonal, each outer product $V_i \otimes K_i^*$ occupies an independent subspace of the $d \times d$ matrix, so retrieval is lossless: $S\tilde{Q} = V_j$ exactly when $K_j^* \cdot \tilde{Q} = 1$ and $K_i^* \cdot \tilde{Q} = 0$ for $i \neq j$. This is in contrast to vector-state models, where superposed bindings in $\mathbb{C}^d$ interfere destructively with retrieval accuracy degrading as $O(1/\sqrt{N})$ [80]. The matrix state supports up to d lossless associations per head; the data-dependent decay then converts this from a lossless store into a controlled lossy one, where information loss is a learned gating decision rather than an unavoidable consequence of the storage format.

During training, the recurrence is computed in $O(T^2)$ time by forming a decay matrix $D \in \mathbb{R}^{T\times T}$ with $\log D[t,i] = \sum_{j=i+1}^{t} \log \gamma_j$ via cumulative sums, applying a causal mask, computing the complex score matrix $W = \tilde{Q} K^{*\top}$, and obtaining the output as $Y = (W \odot D) \cdot V'$. This is mathematically equivalent to the recurrence but parallelizes across the sequence dimension. During autoregressive generation, each token requires $O(Hd^2)$ work per layer, and the state $S \in \mathbb{C}^{H\times d\times d}$ is fixed-size and does not grow with sequence length.

We train and evaluate on WikiText-103 [106], approximately 103 million tokens of Wikipedia text tokenized with the GPT-2 BPE tokenizer (vocabulary size 50,257). The principal results of this work come from a five-scale sweep of the QPAM and RPAM architectures (5M to 100M parameters) under a single canonical training configuration; the per-scale architectural parameters are listed in Table I and the shared training hyperparameters in Table II. All blocks interleave a ComplexGatedUnit (CGU) with a PAM layer, with Gated State Protection enabled, complex RoPE applied to Q and K, no

TABLE I. QPAM scaling sweep architectures.

Scale	dim	layers	heads	head dim
5M	44	12	2	16
10M	80	12	4	16
25M	200	6	4	16
50M	292	12	4	16
100M	384	16	6	64

TABLE II. Canonical training hyperparameters (shared across all scales of the sweep).

Parameter	Value
Optimizer	AdamW
Learning rate	3×10^{-5}
Weight decay	0.01
Warmup steps	500
LR schedule	warmup + cosine decay
Batch size	8
Sequence length	512
Epochs	10
Gradient clipping	1.0
Hardware	Apple M4 Max
Initialization	orthogonal (complex linear maps)

QK phase normalization, and a CGU expansion factor of 3 throughout. RPAM matches each scale's parameter count by using a wider real dimension and additional memory banks (configurations omitted for brevity; see release notes).

Preliminary tests at different hyperparameters, on different hardware (NVIDIA RTX 4090 with bf16 mixed precision and `torch.compile`), and against dense transformer baselines showed qualitatively similar behavior to what we report from the canonical sweep. We restricted the principal computational runs to the configuration above for consistency across model sizes and architecture classes.

Generation samples are logged every 5,000 steps using temperature 1.0, top-k 50, top-p 0.9, and repetition penalty 1.2.

RESULTS

The central question is whether a language model whose representations and operations are entirely complex-valued can learn to model language competitively. If the complex formalism is incompatible with the structure of natural language, or if the conjugate inner product is an ineffective retrieval mechanism, training should fail to converge or produce substantially worse perplexity than a matched real-valued baseline.

The interleaved PAM configuration trains stably on WikiText-103. An earlier sequential configuration (16 CGU layers followed by 16 PAM layers, no RoPE) underperforms it, and a hybrid that adds sparse windowed attention every fourth block produces no improvement, indicating that interleaving channel and sequence mixing matters and that supplemental attention provides no benefit at this scale. A variant with per-element unit normalization of Q and K before the conjugate inner product saw decreasing validation loss but collapsed into severe lexical repetition by mid-training and was stopped during epoch 5, indicating that both magnitude and phase must be free to vary for the retrieval mechanism to function.

A preliminary single-point comparison at ∼100M parameters places PAM within ∼10% of a matched dense transformer baseline. A thorough evaluation against the diversity of contemporary transformer implementations — dense transformers across scales, mixture-of-experts variants, and architectures with alternative attention mechanisms — is left to future work, given the breadth of the design space. We note here only the structural differences: the transformer requires $O(T)$ work per token during inference because of the KV cache, whereas PAM's state is fixed at 49,152 floats per layer regardless of sequence length, giving it $O(1)$ per-token cost. At $T = 2048$, the transformer's KV cache per layer is ∼56× larger than PAM's state, and this ratio grows linearly with context length. Given that the model trains stably and reaches competitive perplexity, the next question is whether the conjugate inner product actually produces the interference the formalism predicts, or whether the complex arithmetic is functioning as a redundant parameterization of a real-valued computation. If the retrieval is effectively real-valued despite the complex representation, $\mathrm{Re}(K^* \cdot Q)$ should be non-negative for all key-query pairs, as it would be in a standard dot-product model. Fig. 1 shows that this is not the case. At layer 0, the distribution of $\mathrm{Re}(K^* \cdot Q)$ vs. $\mathrm{Im}(K^* \cdot Q)$ across all memory banks is nearly symmetric about the origin, with 52% of entries having negative real part — the retrieval is producing destructive interference on a majority of stored associations. By layer 15, the distribution has shifted: the fraction of destructive entries decreases to 40% and the mean of $\mathrm{Re}(K^* \cdot Q)$ moves from -0.8 to $+9.0$. The model learns to make retrieval progressively more selective through depth, using destructive interference to suppress irrelevant associations in early layers and converging toward constructive alignment in later layers.

To make this concrete, Fig. 2 shows the retrieval pattern for two specific sentences at layer 8, averaged across all six memory banks. Each cell shows $\mathrm{Re}(K^* \cdot Q)$ between a query token (row) and a key token (column), with red indicating constructive interference and blue destructive. In "The bat flew out of the cave at dusk," the query at "dusk" retrieves constructively from the prepositional chain "at," "cave," "of" while suppressing "flew" and "bat." In "The bank approved his loan application last week," "loan" retrieves constructively from "approved" and "bank" while "week" suppresses "appli-

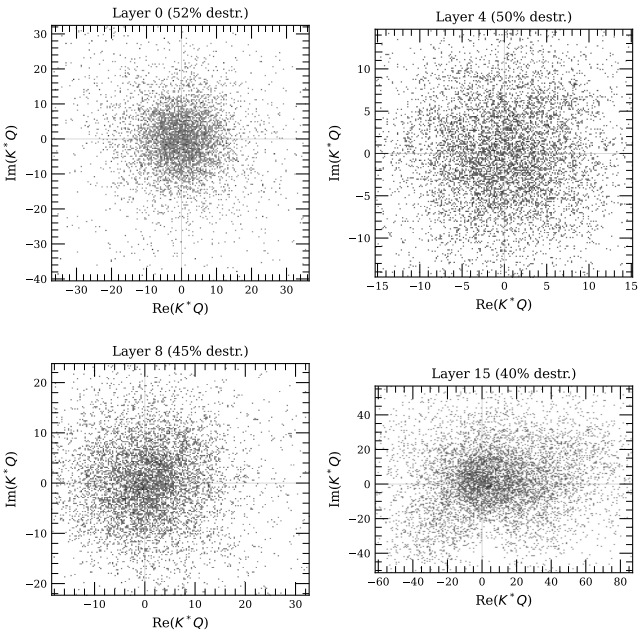

FIG. 1. Distribution of $\mathrm{Re}(K^* \cdot Q)$ vs. $\mathrm{Im}(K^* \cdot Q)$ at layers 0, 4, 8, and 15 across all memory banks. Destructive interference (negative Re) decreases from 52% to 40% through depth; the mean shifts from -0.8 to $+9.0$.

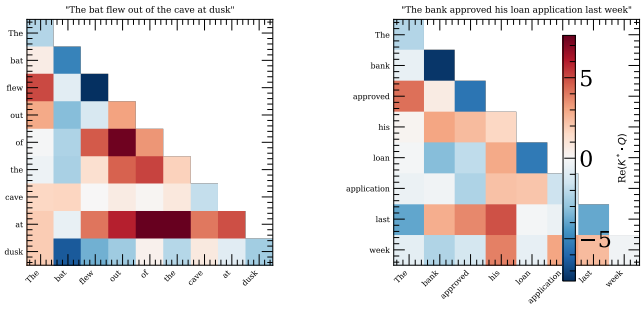

FIG. 2. Conjugate retrieval pattern $\mathrm{Re}(K^* \cdot Q)$ at layer 8 for two sentences, averaged across all six memory banks. Red: constructive interference (retrieved). Blue: destructive (suppressed). The causal mask restricts retrieval to preceding tokens.

cation." These patterns emerge from training on next-token prediction alone.

If the interference patterns are structured at the level of retrieval, the question is whether the underlying representations also reflect this structure. We measured the conjugate inner product $\langle z_1^* | z_2 \rangle$ between learned complex embeddings for 10,256 WordNet synonym pairs and 10,256 random pairs. The real part of the normalized conjugate inner product is significantly higher for synonyms than for random pairs ($d_{\mathrm{sep}} = 0.059$, $p = 1.4 \times 10^{-4}$). Fig. 3 shows a curated subset: synonyms cluster near zero phase difference at elevated coherence, while unrelated pairs scatter across $[-\pi, \pi]$. No supervision over semantic relations was provided during training.

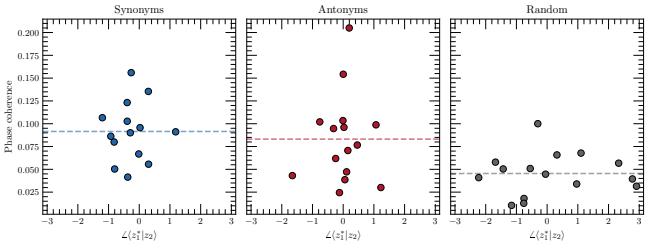

FIG. 3. Phase coherence vs. phase difference $\angle\langle z_1^* | z_2 \rangle$ for synonym, antonym, and random word pairs in the learned complex embeddings. Dashed lines indicate group means ($p = 0.0004$, synonyms vs. random).

TABLE III. Validation perplexity under perturbation of the trained complex embeddings.

Condition	Val PPL
Full complex (baseline)	44.5
Zero imaginary	4,306
Random phase	449,000
Unit magnitude	$> 10^{18}$

The matrix state S_t that accumulates these associations operates well below its theoretical capacity. The effective rank of S_t, measured via the entropy of its singular value distribution, saturates at approximately 10 out of $d = 64$ within the first 10–15 tokens and remains bounded thereafter. The learned decay keeps the state sparse, maintaining $\sim$10 active associations per memory bank at any given time. The d^2 lossless capacity described in Sec. sets the ceiling; the gated decay determines how much of it is occupied at each step.

Perturbing the trained complex embeddings confirms that the model depends on both components of its complex representations (Table III). Zeroing the imaginary component increases perplexity from 44.5 to 4,306; randomizing phase while preserving magnitude yields 449,000; normalizing to unit magnitude while preserving phase causes divergence. These perturbations destroy information the model was trained to use, so the degradation demonstrates dependence, not inherent advantage.

To isolate the contribution of the complex formalism from the matrix-state architecture, we trained a real-valued variant (RPAM) with architecturally identical structure but all complex operations replaced by real-valued equivalents. Because each complex linear map carries two weight matrices (W_r, W_i), RPAM uses a wider dimension and additional memory banks at each scale to match the total parameter count of the corresponding QPAM model. The only difference is the arithmetic: RPAM uses the standard dot product $K_i \cdot Q_t$ for retrieval, real-valued outer products for accumulation, and ReLU-based activations in place of modReLU.

We trained both QPAM and RPAM at five model sizes

spanning 5M to 100M parameters under identical canonical conditions (lr $= 3 \times 10^{-5}$, batch size 8, sequence length 512, 10 epochs, M4 Max). For each trained model we report the mean validation metric over the late-epoch sample window (end-of-epoch and best mid-epoch evaluations from the last three epochs), with standard deviations propagated to log space via $\sigma_{\log_{10} y} = \sigma_y/(y \ln 10)$. Both validation loss and validation perplexity decrease monotonically with parameter count for both models (Fig. 4). Linear regressions in $\log_{10}$–$\log_{10}$ coordinates — fit on loss, the form directly comparable to standard neural scaling laws [43, 44], and on perplexity — give QPAM a loss slope of -0.15 against RPAM's -0.12, and corresponding PPL slopes of -0.65 against -0.49. RPAM has lower absolute loss at every scale we measured (5.56 vs. 4.78 nats at 5M, 3.56 vs. 3.26 at 100M), but the gap narrows monotonically with scale, and the two fits intersect at ∼4.5B parameters in loss space (loss ≈ 2.05 nats) and at ∼550M in PPL space (PPL ≈ 10.6).

The 2.05-nat crossover lies close to the value the Kaplan power law predicts when extrapolated to 4.5B parameters on WebText2 (∼ 2.12 nats [43]), and is within ∼ 0.4 nats of the Kaplan/Chinchilla irreducible-loss estimate of ∼ 1.69 nats [44]. GPT-2 1.5B reaches ∼ 2.3 nats on WebText2 [107], Chinchilla 70B reaches ∼ 1.93 nats [44], and GPT-3 175B reaches ∼ 3.0 nats (PPL 20.5) on Penn Treebank [36].

Assuming the formalisms of the quantum semantic framework [48, 66], in which semantic expressions are represented as states in a complex Hilbert space, the conditional state $\rho_{t|c}$ associated with a context c governs the Born-rule probabilities for the next token t. A real-valued architecture with infinite capacity asymptotes at the Shannon entropy of $\rho_{t|c}$'s diagonal in the discrete-token basis, $H(\operatorname{diag}\rho_{t|c}) = -\sum_t \rho_{tt} \log \rho_{tt}$, since the only outputs it can produce are classical probability vectors over tokens. A complex-valued architecture can in principle represent the full conditional state and asymptotes at its von Neumann entropy, $S_{\mathrm{VN}}(\rho_{t|c}) = -\operatorname{Tr}(\rho_{t|c} \log \rho_{t|c})$. The two are related by the elementary inequality $H(\operatorname{diag}\rho) \geq S_{\mathrm{VN}}(\rho)$, with equality iff ρ is already diagonal in the chosen basis, and the difference $\Delta_{\mathrm{deco}} = H(\operatorname{diag}\rho) - S_{\mathrm{VN}}(\rho) \geq 0$ is the relative entropy of decoherence, the information-theoretic cost of being structurally restricted to the diagonal of a state with off-diagonal coherences. We propose that one can interpret the empirical 1.69-nat real-valued floor as the diagonal projection of the complex-valued von Neumann entropy of $\rho_{t|c}$ onto the classical subalgebra.

Given this framing, it necessarily follows that there should exist some gap in the irreducible loss between complex- and real-valued representations, with the size set by the structure of $\rho_{t|c}$. For a three-state conditional written as the rank-one projector onto a maximally-coherent superposition with non-trivial phases between basis vectors,

$$\rho_{t|c} \;=\; \frac{1}{3}\begin{pmatrix} 1 & e^{-i\pi/3} & e^{-i2\pi/3} \\ e^{i\pi/3} & 1 & e^{-i\pi/3} \\ e^{i2\pi/3} & e^{i\pi/3} & 1 \end{pmatrix}, \tag{8}$$

the projector onto $|\psi\rangle = \frac{1}{\sqrt{3}}(|0\rangle + e^{i\pi/3}|1\rangle + e^{i2\pi/3}|2\rangle)$. The diagonal is uniform, $\operatorname{diag}\rho_{t|c} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, giving $H(\operatorname{diag}\rho) = \log 3 \approx 1.099$ nats — the maximum Shannon entropy a real-valued architecture can assign to a three-outcome distribution. But $\rho_{t|c}$ is rank one, so $S_{\mathrm{VN}}(\rho) = 0$: a complex-valued architecture that represents the full state assigns it zero entropy. The decoherence gap is the entire $\log 3$, and the whole prediction loss the real-valued architecture incurs for this state is information that lives in the off-diagonal phase coherences. Mixing this state with the maximally mixed one, $\rho(p) = (1-p)|\psi\rangle\langle\psi| + p\mathbb{1}/3$, gives eigenvalues $(1-2p/3,\, p/3,\, p/3)$, leaves the diagonal uniform, and interpolates the gap monotonically from $\log 3$ at $p = 0$ to 0 at $p = 1$. More generally, the maximum decoherence gap for a d-outcome conditional is $\log d$, so anchoring at the empirical $E_{\mathrm{real}}^{\mathrm{Kaplan}} \approx 1.69$ nats per token gives

$$E_{\mathrm{complex}}^{\mathrm{floor}} \;\geq\; E_{\mathrm{real}}^{\mathrm{Kaplan}} - \log d_{\mathrm{eff}}, \tag{9}$$

where d_{eff} is the effective coherence dimension of the conditional state per token-prediction event: $d_{\mathrm{eff}} = 2$ gives a predicted floor of ∼ 1.00 nat per token, $d_{\mathrm{eff}} = 4$ gives ∼ 0.30 nat. The strict positivity of the gap follows from QSF self-consistency alone, and the substantive prediction is that the 1.69-nat figure in scaling-law analyses of real-valued transformers is not a fundamental floor on language modeling but the diagonal projection of a von Neumann entropy that a natively complex-valued architecture is in principle able to reach.

The architectural overhead that handicaps QPAM at small scale — the complex parameterization requires sufficient capacity to stabilize the conjugate-inner-product retrieval — thus becomes a return on investment as scale grows, the pattern one would expect if the real-valued model is approximating, with classical machinery, a structure that the complex-valued model represents natively. With more careful optimization of the complex implementation (custom CUDA kernels, learning-rate schedules tuned for the modReLU activation) and training on corpora large enough that the upper end of the sweep stays within compute-optimal bounds — the 50M and 100M points sit past Chinchilla-optimal for WikiText-103's ∼118M tokens even allowing for multi-epoch repetition — we would expect the crossover transition to occur at smaller N than this extrapolation predicts.

At the 100M point of the sweep (PAM dim 384 with 6 memory banks, RPAM dim 576 with 9), the per-epoch convergence trajectories (Fig. 5) show RPAM stabilizing

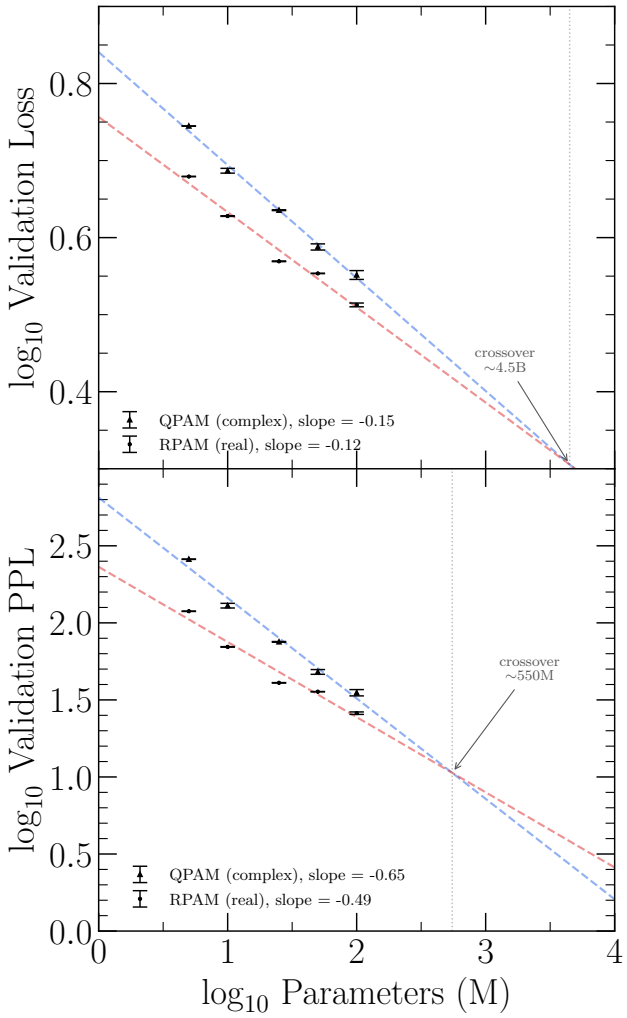

FIG. 4. Scaling of validation loss (top) and validation perplexity (bottom) for QPAM (triangles) and RPAM (stars) on WikiText-103, in $\log_{10}$–$\log_{10}$ coordinates with shared x-axis. Each marker is the mean of late-epoch samples for one trained model; vertical bars are first-order propagated standard errors $\sigma_{\log_{10} y} = \sigma_y/(y \ln 10)$. Dashed lines are linear fits in log–log space, with slopes given in each panel's legend. Dotted vertical lines mark the extrapolated intersection of the two fits in each space: the loss-space fit crosses at ∼4.5B parameters (loss ≈ 2.05 nats), the PPL-space fit at ∼550M parameters (PPL ≈ 10.6).

earlier than QPAM, consistent with the maturity of real-valued optimization and the absence of custom kernels in the complex implementation.

To probe whether the differential scaling in loss and perplexity reflects underlying differences in how the two architectures organize their representational space, we measured the separability between WordNet synonym pairs and random word pairs in the learned token embeddings across the canonical sweep (Fig. 6). For QPAM we used the real part of the normalized conjugate inner product $\mathrm{Re}\langle z_1^* | z_2 \rangle$, and for RPAM the cosine similarity, in each case aggregated over 10,256 synonym pairs and 10,256 random pairs constructed from the GPT-2 single-token portion of the WordNet lemma vocabulary. We report Cohen's d between the two distributions with bootstrap error bars. QPAM's separability traverses zero between 10M and 50M parameters, reaching $d_{\mathrm{sep}} = +0.054 \pm 0.014$ at 50M (Mann–Whitney $p = 8 \times 10^{-4}$, the only point at which either architecture exceeds the random baseline at significance in the canonical sweep). RPAM's separability is negative across the sweep and trends monotonically from $d_{\mathrm{sep}} = -0.116$ at 5M toward zero, reaching -0.019 ± 0.014 at 50M without crossing into positive territory. Both architectures begin below zero, indicating that synonym pairs are slightly less similar to each other than random pairs at the smaller scales, an artifact we attribute to lexical-frequency co-statistics that the small models have not yet organized through. The crossing point for QPAM emerges within the canonical sweep while RPAM remains in the trending-toward-zero regime, consistent with the loss-curve interpretation that the complex-valued architecture organizes representational structure earlier in scale.

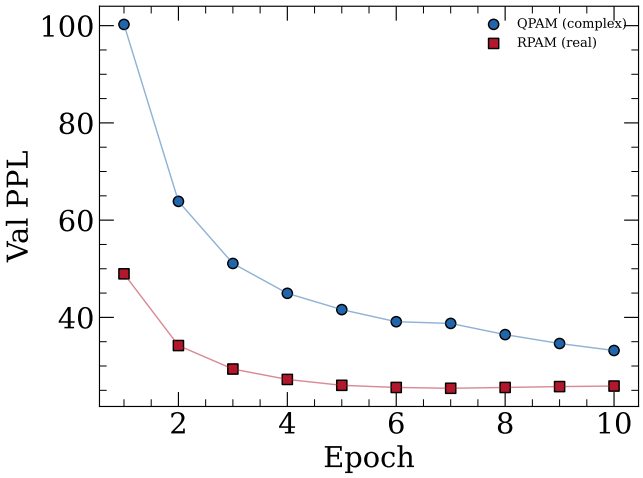

FIG. 5. Per-epoch validation perplexity at the 100M point of the QPAM/RPAM sweep on WikiText-103 (seq_len 512, M4 Max). RPAM converges faster and stabilizes lower at this scale (25.42 vs. 33.19); the scaling-law view in Fig. 4 shows that the gap narrows with model size.

At epoch 10, given the prompt "In 1923, the University of," the model generates: "In 1923, the University of Illinois at Urbana @-@ Urdu said it was 'an easy choice to do something in its own right.' The university also claimed the first students from Wisconsin had to be replaced by a more 'good student' due to a lack of funds." The text is grammatically coherent and shows structural awareness of dates, proper nouns, and institutional language, with 3-gram repetition rate 0.034, 4-gram repetition rate 0.011, and unique token ratio 0.703. Factual accuracy is

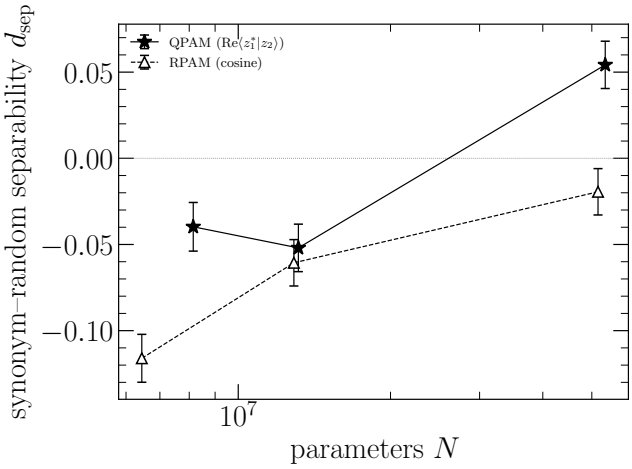

FIG. 6. Synonym–random separability d_{sep} (Cohen's d) of word-pair similarity distributions for QPAM (filled stars, $\text{Re}\langle z_1^*|z_2\rangle$) and RPAM (open triangles, cosine similarity) across the canonical sweep, computed over 10,256 WordNet synonym pairs and 10,256 random pairs at each scale, with bootstrap error bars on d_{sep}. QPAM crosses zero between 10M and 50M parameters and is significant at 50M ($p = 8 \times 10^{-4}$); RPAM trends toward zero from below but does not cross within the canonical range.

not reliable at this scale.

DISCUSSION

Schlag *et al.* [88] showed that linear attention without softmax is equivalent to a fast weight programmer [87] in which outer products of keys and values are accumulated into a matrix that queries retrieve from via matrix-vector product. Subsequent work has refined this structure with exponential decay [90], data-dependent gating [91], delta-rule updates [92], and complex-valued gates [93], while from the LSTM lineage Beck *et al.* [98] arrived at the same recurrence independently. PAM belongs to this family, but the substitution of the conjugate inner product for the standard dot product introduces a qualitative difference that is worth examining in the context of what is known about the limitations of linear attention.

Arora *et al.* [109] demonstrated that linear attention models struggle with associative recall, the ability to retrieve a specific stored value given a matching key, because the non-negative nature of real-valued inner products means that all stored associations contribute positively to the retrieval, diluting the target signal. Ramsauer *et al.* [84] showed that softmax attention avoids this problem because it implements a modern Hopfield network [85, 86] in which the exponential nonlinearity sharpens retrieval to a near-one-hot selection. PAM's conjugate inner product provides a different mechanism for addressing the same problem: because the complex dot product $K_i^* \cdot \tilde{Q}_t$ can be negative or imaginary depending on the phase relationship between the stored key and the query, associations that are phase-incoherent with the query are actively suppressed rather than merely downweighted. Whether this destructive interference is as effective as the exponential sharpening of softmax at recovering specific associations from a large store is a question that passkey retrieval experiments at long context lengths would address directly.

Danihelka *et al.* [83] previously incorporated complex-valued holographic representations into an LSTM and found improvements on tasks requiring compositional binding and retrieval, consistent with the classical result of Oppenheim and Lim [72] that phase carries more structural information than magnitude. The connection between phase and representational capacity has a deeper grounding in the geometric phase of Berry [74] and Pancharatnam [73], which showed that phase relationships encode topological properties of the trajectory a system follows through its state space. In the context of PAM, this suggests that the phase of the complex state may encode not only the content of stored associations but something about the path through which they were accumulated, a form of sequential information that real-valued states cannot represent without additional mechanisms. The QK phase normalization ablation described in , in which removing the magnitude degree of freedom from queries and keys caused generation to collapse into repetition, provides indirect evidence that both magnitude and phase are carrying distinct and necessary information in the retrieval process.

Lo *et al.* [67] used a sheaf-theoretic framework [68] to identify over 77,000 instances of contextuality in BERT's internal representations, demonstrating that the non-separability observed behaviorally in Bell violation experiments [48, 60, 64, 66] is also present in the geometry of the embeddings themselves. This finding connects to the difficulties documented in the mechanistic interpretability literature, where sparse autoencoders trained on frontier models lose approximately 90% of the model's capability when their reconstructions replace the original activations [51], extracted features are neither selective nor independent when used for steering [52], and there is a proven exponential gap between representing features in superposition and computing with them [54]. Williams *et al.* [69] have argued that these difficulties may reflect conceptual limitations of the decomposition framework rather than technical ones. If the representations are fundamentally non-separable, as the Bell violations and the sheaf-theoretic analysis suggest, then an architecture that operates natively in complex Hilbert space may not merely provide an alternative computational mechanism but may produce representations whose structure is more naturally described by the formalism in which they were computed. Whether PAM's representations are in fact more interpretable in sheaf-theoretic terms than transformer representations is a testable question that we in-

tend to pursue.

The observation that the state matrix S_t operates at effective rank ∼10 out of $d = 64$, despite having d^2 lossless capacity available, has a natural interpretation in the complex formalism. The state carries a superposition of all associations accumulated up to time t, and the data-dependent decay selectively suppresses associations that are not reinforced by incoming tokens. Mathematically, a $d \times d$ complex matrix at rank 10 has had most of its off-diagonal coherences suppressed; in the language of quantum information, this is decoherence [110]. The model is not doing quantum physics, but it is using the same algebraic structure for a functionally analogous reason: maintaining a superposition of stored associations is useful for flexible retrieval, but the system cannot act on all of them simultaneously, so it learns to collapse the superposition to the associations relevant to the current context. This is the problem of relevance realization [49] implemented in the mathematics of the complex Hilbert space that operational quantum logic identifies as native to contextual systems. The decay is the decoherence mechanism; the conjugate retrieval is the measurement.

PAM's matrix state $S_t \in \mathbb{C}^{d\times d}$ is also an operator in the sense of the neural operator framework of Kovachki *et al.* [111], which formalizes learned maps between function spaces. The universal approximation theorem for operators [112], recently extended to show that nonlocality and nonlinearity suffice for universality [113], applies to PAM's conjugate retrieval: the complex-valued kernel $K(q, k) = q^* \cdot k$ is both nonlocal (all stored associations contribute) and nonlinear (through the decay and gating mechanisms). The kernel interpretation of attention developed by Tsai *et al.* [114] connects the discrete retrieval to a continuous integral operator whose selectivity is phase-dependent. Work on data encoding in variational quantum models [115] has shown that encoding classical data into quantum Hilbert space can provide expressivity equivalent to infinitely wide classical networks; PAM's complex embedding can be understood as a classical implementation of this encoding. The Fourier neural operator [116], which learns in the spectral domain using inherently complex-valued Fourier coefficients, provides an independent demonstration that complex arithmetic in operator learning captures structure efficiently. Characterizing the operator S_t itself — its eigenspectrum, the diversity of operators the model produces across different inputs, and whether the operator vocabulary changes with sampling temperature — is a natural direction for future analysis.

The computational costs are comparable in order to standard attention: the training-time dual form gives $O(T^2Hd)$ per layer, while at inference PAM requires $O(Hd^2)$ per token using a fixed state of 49,152 floats per layer regardless of sequence length, compared to a KV cache that grows linearly with context. The throughput gap we observed in preliminary GPU tests against a matched transformer reflects the arithmetic cost of complex linear maps and the absence of custom CUDA kernels rather than a fundamental asymmetry in computational complexity.

CONCLUSION

In this work, we have used a five-scale parameter sweep on WikiText-103 (5M to 100M parameters) to investigate the scaling behavior of Phase-Associative Memory (PAM), a language model whose representations and operations are entirely complex-valued, in comparison with a matched real-valued ablation (RPAM). Our conclusions are the following:

1. Across the five-scale sweep under identical training, PAM and RPAM both show validation perplexity decreasing monotonically with parameter count (Section). Linear fits in log–log space [43, 44] give PAM a slope of -0.65 and RPAM a slope of -0.49. RPAM has lower absolute perplexity throughout the measured range, but the gap narrows monotonically with scale (from 2.18× at 5M to 1.36× at 100M) and the extrapolated fits intersect at ∼550M parameters. The slope differential suggests that the complex formalism returns more compression per additional parameter than the real-valued ablation.

2. Under the formalisms of the quantum semantic framework [48, 66], the empirical ∼1.69 nat irreducible-loss floor characterized for real-valued transformer fits [43, 44] can be interpreted as the diagonal projection of the complex-valued von Neumann entropy of the conditional state $\rho_{t|c}$. Within this framing, it necessarily follows that there should exist some gap in the irreducible loss between complex- and real-valued representations, with the size set by the structure of $\rho_{t|c}$, and that a natively complex-valued architecture is in principle able to reach below the real-valued floor.

3. Phase-Associative Memory accumulates associations in a complex matrix state via outer products and retrieves via the conjugate inner product, and trains stably on WikiText-103 with the matrix-state recurrence and complex parameterization carrying the entire computational signal path without recourse to softmax attention. The qualitative retrieval mechanism — conjugate inner product with destructive interference — is one the real-valued ablation cannot produce.

4. Holographic binding in vector-state models fails due to the $O(1/\sqrt{n})$ capacity degradation of superposed associations. Upgrading the state from $\mathbb{C}^d$ to $\mathbb{C}^{d\times d}$ resolves this, consistent with the classical

analyses of superposition capacity in holographic reduced representations [80].

5. Both magnitude and phase must be free to vary for the conjugate inner product to function as a retrieval mechanism; normalizing queries and keys to unit magnitude causes generation to collapse into repetition.

6. Supplementing PAM with sparse windowed attention provides no benefit at the scales we explored, suggesting that the information attention would contribute is already being captured by the complex recurrence.

7. PAM's retrieval operation implements the same mathematical structure as the quantum semantic framework [48]: outer-product accumulation of associations in a complex Hilbert space and projection of a query onto the accumulated state through the conjugate inner product, the operation that computes measurement probabilities in quantum mechanics.

8. The synonym–random separability d_{sep} of the learned token embeddings (Fig. 6) crosses zero for QPAM between 10M and 50M parameters and reaches $+0.054 \pm 0.014$ at 50M ($p = 8 \times 10^{-4}$), while RPAM's d_{sep} trends toward zero from below across the canonical sweep without crossing, consistent with the loss-curve interpretation that the complex-valued architecture organizes representational structure earlier in scale.

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